6.7 Integrals Containing Trigonometric Functions

Name _____ Period ____ Date _____

AP Calculus 6.7 · Student sheet

I can…

- integrate odd powers of sine and cosine.
- use power-reducing identities for even powers.
- use identities for tangent and secant and for products like sin(ax)cos(bx).

Vocabulary · fill in the missing word

Pythagorean identity — $\sin^2 x + \cos^2 x =$ _____

Power-reducing identities — $\sin^2 x =$ _____ $= \dfrac{1 + \cos 2x}{2}$

Product-to-sum identity — $2 \sin A \cos B =$ _____

Part 1 · Which u?

$\int \sin x \cos^2 x \, dx$

Try	du	Does it work?
$u = \sin x$		
$u = \cos x$		

Part 2 · Notes and examples (fill in each note, then work the example)

Odd power of sine: save one sin x and change the rest with $\sin^2 x = 1 -$ _____; then let $u = \cos x$.

Example 1

Find $\int \sin^3 x \, dx$.

Even powers: use the _____ identities.

Example 2

Find $\int \cos^2 x \, dx$.

Tangent: $\tan^2 x =$ _____ $- 1$.

Example 3

Find $\int \tan^2 x \, dx$.

Products: $\sin(ax)\cos(bx) = (1/2)[\sin((a - b)x) + \sin((a + b)x)]$, the _____ identity.

Example 4

Find $\int \sin(3x)\cos(2x)\,dx$.

Example 5

Evaluate $\int$ from 0 to π of $\sin^2 x\,dx$.

Part 3 · Partner practice

Partner 1

Find $\int \sin^2 x \cos x\,dx$.

Partner 2

(AP® Practice) $\int$ from 0 to $\frac{\pi}{2}$ of $\cos^2 x\,dx =$

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{1}{2}$ (D) 1